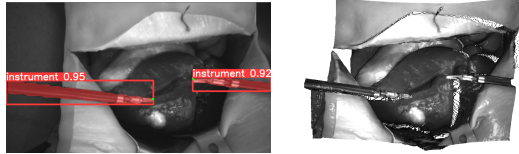


Figure 4: Exemplary online measurement in the Playground.



Original left image.

Point cloud.

Figure 5: Exemplary reconstruction of the Phantom and instruments.

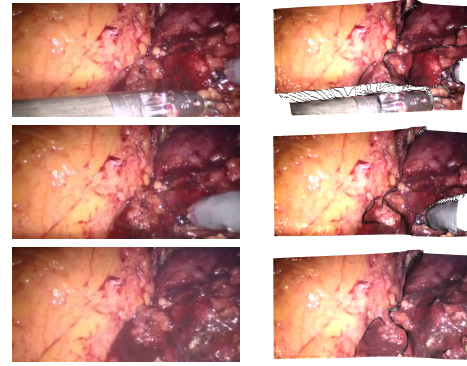
tion to the medical workflow. However, predicting the adoption of such a method by surgeons in general is challenging (?). This is exacerbated by the lack of a comparable system and data on usage frequency. An additional point of consideration is the optimal measurement process in practical settings. Currently, accurate measurements require that both surgical instruments, including their tooltips, remain visible within the camera’s field of view.

Path to Deployment

Addressing the limitations mentioned above, several research directions can enhance our method. First, it is worth considering the enhancement of the mesh creation component and optimizing hyperparameters. Exploring AI models for generating a mesh from point clouds is promising (?). Similar to RAFT-Stereo, the AI model could recognize structures and thus represent a more intelligent approach than PSR. Our method’s modularity allows for the optimization or replacement of elements in future studies. Second, it would be of interest to refine the prototype by using a laparoscopic stereo camera and involving surgeons in real-world measurements (?). In addition, we consider methodologies that map multiple reconstructions of internal structures, allowing for persistent reference points even when they transition out of the camera’s field of view. Collaborating with surgeons will pinpoint weaknesses and possible applications. Additionally, integrating workflow recognition could enable automatic measurement of actions like cuts. Lastly, as reliable measurement accuracy is crucial, integrating measurement uncertainties by estimating error margins is an important future research avenue.

Conclusion

This work aims to develop a versatile measurement tool for laparoscopy. We outline the fundamental characteristics of laparoscopy and establish the key requirements for a laparoscopic measurement tool. Based on these requirements, we



Original left image.

Point cloud.

Figure 6: Exemplary point clouds of real surgery images.

devise a method that intelligently integrates the results of existing research into a formally defined framework.

We evaluate our method through a series of experiments designed to test various functionalities. To achieve this, we implement the method as a prototype, integrating several state-of-the-art components and establishing appropriate experimental environments. For distance measurements, we design and 3D print a CAD model, while a surgical phantom serves for more realistic application scenarios. We construct datasets for both environments, opting for a challenging setting to evaluate the robustness of our method. In addition, we supplement these two datasets with a third public dataset comprising real-world laparoscopic images.

The results show the potential of our proposed human-AI-based measurement method. In addition to exhibiting a remarkably high accuracy, it demonstrates substantial robustness when applied to real-world image data. The inclusion of AI-based components, such as RAFT-Stereo and YOLOv8, is instrumental in achieving these results. Moreover, we successfully show the essential functionality of online measurements by employing surgical instruments to establish reference points. Although the measurement outcomes are not as accurate as those in the offline selection, they still indicate potential. Considering that we do not optimize the individual components for laparoscopic application in the current setting, these results serve as a proof of concept.

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